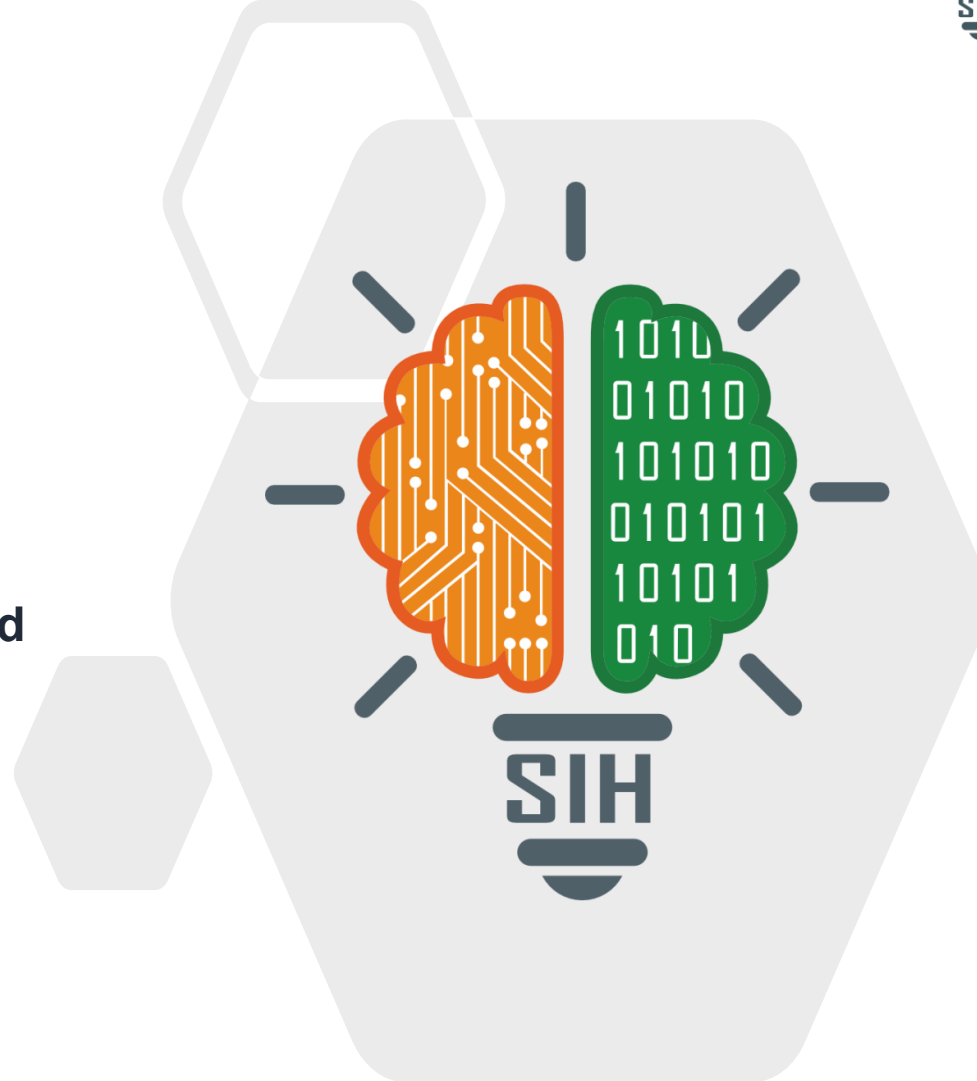


# SMART INDIA HACKATHON 2026



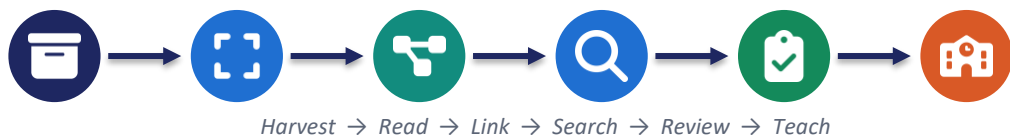
- **Problem Statement ID : 26063**
- **Problem Statement Title : Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal**
- **Theme : Smart Education**
- **PS Category : Software**
- **Team ID :**
- **Team Name : Anarch**



# Polar Knowledge Index

“Every expedition. Every page. **Every claim cited.**”

- Proposed Solution:** A knowledge index built on what NCPOR already publishes — harvested, linked to each expedition, and cited to the page.



## What it does?

- Harvests, never re-enters** : DSpace expedition reports (OAI-PMH) + NCPOR news feed, polite & cached — originals stay where they are
- Reads the scans** : PDF text layer or Tesseract OCR, page image kept for “view original page”
- Links everything** : item → expedition · station · topic · year; machine links stay unconfirmed until a curator confirms

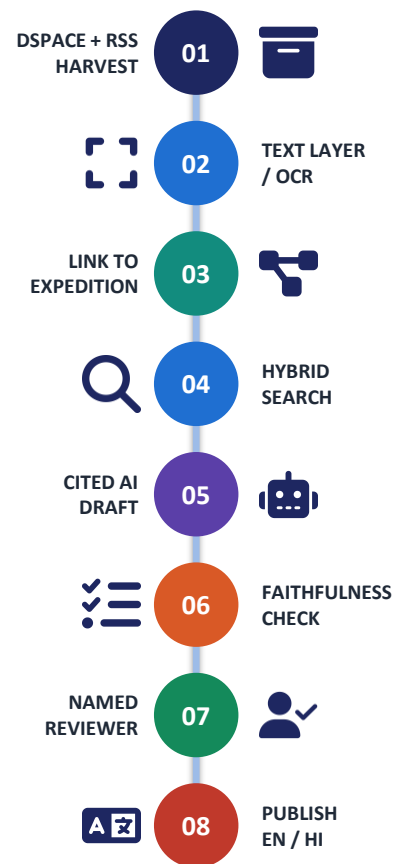
## How it answers the PS?

- Repository** : 32 expedition reports, 765 papers and 837 news posts in one index, one page per expedition
- Dissemination** : cited answers, lesson units and web + social posts, in English and Hindi

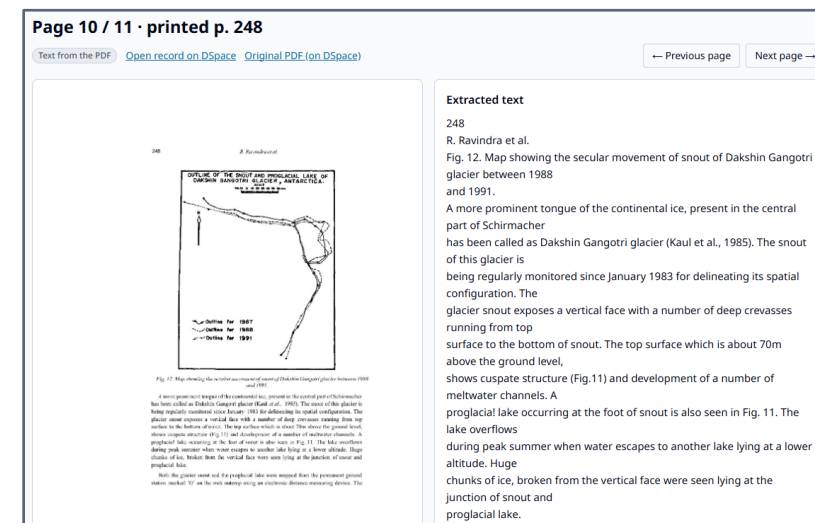
## What is different about it?

- Real NCPOR archive** , zero new data entry for staff
- Checked, not claimed** : every sentence fact-checked against its page; unsupported ones block approval
- Named reviewer** + append-only audit log; refuses when the archive has no source
- Coverage view** : shows which expeditions have no public report or story

## WORKFLOW



## PROTOTYPE IMAGE



Working prototype · real DSpace scan · 9th Expedition glacier paper, printed p. 248 — scan beside its searchable text

**Scan** : the page exactly as NCPOR published it

**Citation** : every answer links back to this page

**Coverage** : expeditions with no public report

github repo link : <https://github.com/saransh2809/SIH26063-Polar-Knowledge-Index>  
demo youtube video :



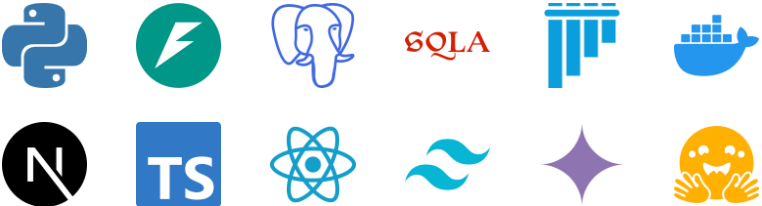
# TECHNICAL APPROACH



Technology stack & system architecture — built and running end-to-end on NCPOR’s real archive

## Technology Stack

| LAYER     | TOOL                                                       |
|-----------|------------------------------------------------------------|
| Harvest   | httpx · OAI-PMH · lxml · robots.txt-aware, 1 req/s, cached |
| Documents | PyMuPDF · Tesseract OCR 5.4 · page images                  |
| Search    | PostgreSQL 16 full-text + pgvector · multilingual-e5-base  |
| AI        | Google Gemini — grounded drafting, checker, Hindi          |
| Backend   | Python 3.12 · FastAPI · SQLAlchemy 2 · Alembic             |
| Frontend  | Next.js · TypeScript · Tailwind · WCAG 2.1 AA              |
| Security  | Staff roles · JWT · bcrypt · append-only audit             |
| Quality   | 25 automated tests (pytest) · Docker Compose               |



- ✓ Originals never re-hosted: links + OCR text + page images only
- ✓ Public is read-only; every staff action is audited

## PIPELINE



**Evaluation:** hand-verified test set, scored once → citation accuracy [\_\_%] · refusal accuracy [\_\_%]



# FEASIBILITY AND VIABILITY



✓ FEASIBILITY — already proven, not projected

➔ VIABILITY — path to operations



797

reports & papers  
harvested, zero entry



3,266

scanned pages  
read & searchable



52

expeditions in  
the coverage view



25

automated tests  
passing

- **Runs on a laptop CPU:** search model runs locally — no GPU; one database, one backend
- **Zero new work for NCPOR:** harvests DSpace + news that already exist; review is the only staff task
- **Open standards, open source:** OAI-PMH, PostgreSQL, Next.js — zero licence cost

## ⚠ RISKS

## 🛡 HOW WE HANDLE THEM

1980s scans read badly

Text layer first; OCR confidence shown; original page beside every quote

AI states what the source doesn't

Per-sentence checker; unsupported blocks approval; refusal threshold

Looks like other SIH AI portals

Real archive, zero entry, coverage view, measured accuracy

NPDC not open to harvesting

Link-only until NCPOR permits; adapter slot ready

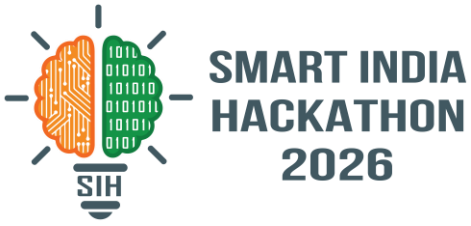
AI API outage or rate limits

Retries + cached answers; search works without AI





# IMPACT AND BENEFITS



## PRACTICAL IMPACT

Case study • a student asks: “Has India measured how the Dakshin Gangotri glacier is moving?”

① Search finds the papers

② Cited draft, every sentence checked

③ Click through to the scan

Answer drawn only from glaciology papers in the 9th, 11th, 26th and 30th Expedition reports — e.g. the snout **receded 81 cm in 2011** — each sentence links to its page. Off-archive questions are **refused**.

## QUANTIFIED IMPACT

6 → 1

disconnected systems become one index

0

new records typed by NCPOR staff

[ \_ % ]

citation accuracy, hand-verified

## WHO BENEFITS ?

NCPOR outreach & MoES

Teachers & students

Researchers & PhD scholars

Journalists & media

Polar & Ocean Museum

## BENEFITS OF THE SOLUTION

SOCIAL

Students get answers they can trust, cited to India’s own reports

ECONOMIC

No new data-entry cost; open-source stack, zero licence fees

OPERATIONAL

Outreach posts drafted in minutes from confirmed records

TECHNOLOGICAL

Every public sentence traceable to a page, with an audit log

ENVIRONMENTAL

Glacier, ice and climate evidence reaches classrooms

SCIENTIFIC

40 years of expedition science searchable; fewer duplicated baselines

NATIONAL

India’s polar programme taught with Indian data; Hindi + GIGW 3.0



# RESEARCH AND REFERENCES



## Key research findings (built on NCPOR’s real archive)



### The archive exists — the links don’t

Six separate NCPOR systems; DSpace already speaks OAI-PMH, so 32 reports and 765 papers were harvested with no staff effort



### Old scans are more readable than feared

9 of 10 reports carry a usable text layer on 68–92% of pages; the 30th was fully OCR’d at 90% mean confidence



### AI must be checked sentence by sentence

In our glacier lesson test the checker flagged 4 of 27 checked sentences unsupported — incl. a general “glaciers move slowly” line absent from the sources

## References :

- [DSpace at NCAOR — Scientific Reports of Indian Expeditions to Antarctica](#)
- [NCPOR news feed \(RSS\)](#)
- [National Polar Data Center \(NPDC\), NCPOR](#)
- [OAI-PMH v2.0 — Open Archives Initiative protocol](#)
- [Wang et al. \(2024\) — Multilingual E5 text embeddings, arXiv](#)
- [Cormack, Clarke & Buettcher \(2009\) — Reciprocal Rank Fusion, SIGIR](#)
- [Lewis et al. \(2020\) — Retrieval-Augmented Generation, NeurIPS](#)
- [WCAG 2.1 — W3C accessibility guidelines \(GIGW 3.0 basis\)](#)

github repo link : <https://github.com/saransh2809/SIH26063-Polar-Knowledge-Index>  
demo youtube video :

## Comparison with existing approaches

| Capability                                | NCPOR today | Typical SIH prototype | Polar Index |
|-------------------------------------------|-------------|-----------------------|-------------|
| Uses NCPOR’s real archive, no re-entry    | ✗           | ✗                     | ✓           |
| Links every item to its expedition        | ✗           | partly                | ✓           |
| Page-level citation for AI text           | ✗           | ✓                     | ✓           |
| Sentence-level fact-check blocks approval | ✗           | ✗                     | ✓           |
| Named reviewer + append-only audit        | ✗           | partly                | ✓           |
| Measured accuracy on verified set         | ✗           | ✗                     | ✓           |
| Coverage view of programme gaps           | ✗           | ✗                     | ✓           |
| Hindi output, reviewed                    | ✗           | ✓                     | ✓           |

“Typical SIH prototype” = public repositories for PS SIH26063 reviewed before building (e.g. PolarCROSS, PRISM, POLARIS, DhruvGyani). NCPOR today = main site, DSpace, NPDC, MET-Data, library and social channels as separate systems.

**Coverage view finding:** reports for the 7th and 8th expeditions are listed but unpublished; no DSpace report exists after the 30th — 12 later expeditions (33–37, 40–46) are known only from NCPOR news.